

Machine Learning Approaches to Online Transaction Fraud Detection

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Section 1: Background and Objective

With the rapid growth of e-commerce and digital payment systems, transaction fraud has become a crucial concern for financial institutions, since undetected fraudulent transactions may lead to financial loss, operational disruption, reputational damage, and regulatory consequences. Given the large volume and speed of transaction processing, suspicious transactions are often difficult to identify through manual review alone, resulting in a need for data-driven fraud detection tools.

Consequently, we conducted a mini case study to evaluate the efficacy of leveraging machine learning models for fraud detection. This case study aims to develop a fraud detection framework for predicting whether an online transaction is fraudulent using the provided dataset. Building on the exploratory data analysis, this study focuses on the modelling and evaluation stages by developing two tree-based classification models, namely the Decision Tree and the Random Forest, for fraud identification.

The study has three main objectives. First, it aims to construct and compare the predictive performance of the Decision Tree and Random Forest models in detecting fraudulent online transactions. Second, it aims to interpret the models by identifying the features that contribute most to the discrimination between fraudulent and non-fraudulent cases, thereby improving the accountability and practical usefulness of the fraud detection system. Third, it provides an analytical basis for the later discussion of non-data analytic elements, such as broader fraud risks, internal controls, and other preventive considerations beyond model performance. As the main priority of this case study is to capture fraud cases, greater emphasis is placed on recall, since failing to detect a fraudulent transaction (False Negative) is generally more costly than incorrectly flagging a legitimate one (False Positive).

Section 2: Exploratory Data Analysis (EDA)

Step 1 - Distinguish Attributes:

Procedure: Executed the `str(*)` and `summary(*)` to analyze the attribute of the dataset.

Findings: Based on the structure table (**Appx 1**), the dataset contains 100000 observations and 101 columns, comprising 100 feature variables and 1 target variable labeled “*isFraud*”. Among the 100 feature variables, 19 are character-type, 58 are integer-type, 14 are logical-type, and 9 are numeric-type. Then, we grouped the variables into 67 numerical features (integer and numeric types) and 33 categorical features (character and logical types). The summary table (**Appx 2**) shows missing values in certain variables. In addition, several features exhibit disproportionately large maximum values relative to their means, suggesting the presence of potential outliers.

Step 2 - Missing and Constant Value Handling:

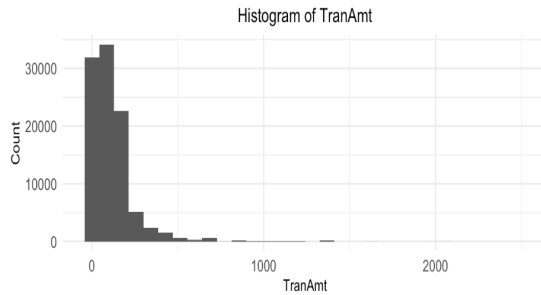
Procedure: Calculated the missing rate for each feature and then applied simple imputation strategies based on data type. Numerical features were imputed using the median to reduce the influence of extreme values, while categorical features were imputed with a constant value (“miss”) to capture missingness as a separate category. Also, we removed variables that were either constant (single-value across all observations) or entirely missing, since they provide no discriminatory power for fraud analysis.

Findings: Based on the missing ratio table (**Appx 3**), a total of 10 variables were identified as entirely missing, and 2 variables were found to be constant across all observations. All of them were removed.

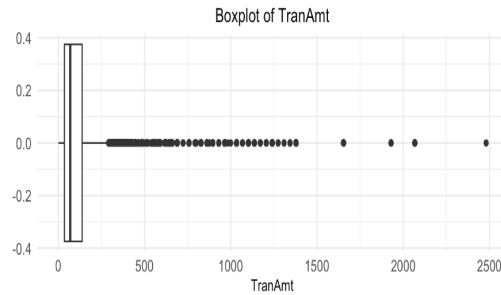
Step 3 - Univariate Analysis:

Procedure: Plotted the histogram & boxplots for every numerical feature and plotted the bar chart for every categorical feature.

Findings:

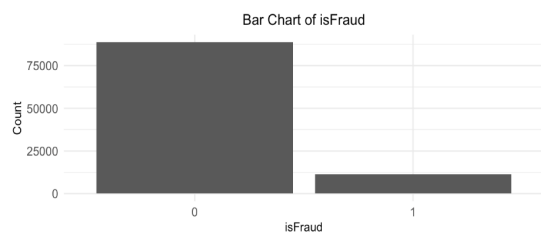


(Fig. 1: Histogram of TranAmt)

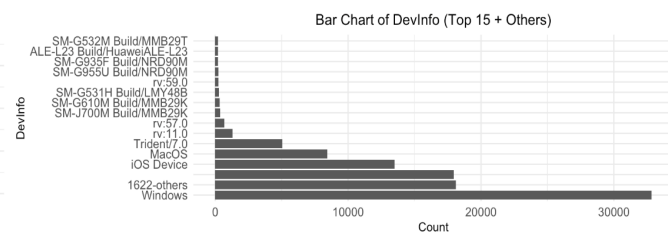


(Fig. 2: Box plot of TranAmt)

Based on the histograms (**Appx 4**) and boxplots (**Appx 5**) of all numerical features, most variables show skewed distributions, implying the presence of many outliers. Taking the feature “TranAmt” as an example, **figure 1** shows that most transaction amounts are concentrated at lower values, while only a small number occur at higher values. The boxplot in **figure 2** further confirms this pattern, with the box positioned toward the lower-value range and several outlier points appearing at higher values. This suggests that the dataset contains many outliers that should be addressed.



(Fig. 3: Bar of isFraud)



(Fig. 4: Bar of Devinfo)

Furthermore, based on the bar charts (**Appx 6**) of label and all categorical features, the distributions appear imbalanced, and some categorical features contain a large number of unique values. **Figure 3** shows that the number of cases labeled 0 in “isFraud” is much higher than those labeled 1, indicating that the dataset may be imbalanced. **Figure 4** illustrates the example of “devInfo”, which contains more than 1600 values, as shown in “1622-others”. Thus, it should be transformed using binning.

Step 4 - Outlier Handling:

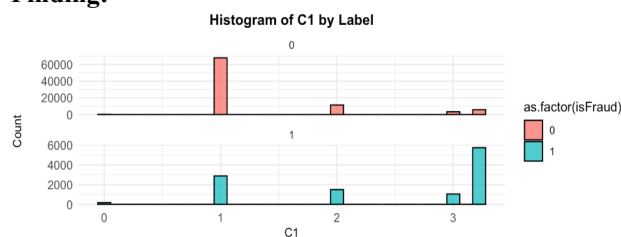
Procedure: Following Van Gestel and Baesens method, outliers were identified using the threshold $M \pm \frac{3 \cdot IQR}{2 \cdot 0.6745}$. As extreme values may indicate fraudulent behaviour, conditional winsorization was applied to cap observations at the calculated bounds. Winsorization was performed only when feature variability was sufficient ($IQR \neq 0$) and when the proportion of affected observations exceeded 0.1%, preserving meaningful outliers while limiting the influence of extreme values. Hence, the outlier rate for each feature and the overall average outlier rate were calculated.

Findings: Based on the outlier rate table (**Appx 7**), the average outlier rate is 11.18%, indicating the presence of a considerable proportion of extreme observations across the dataset.

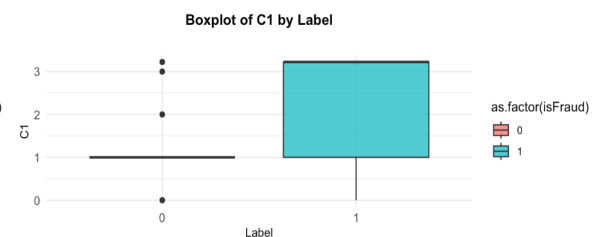
Step 5 - Bivariate Analysis:

Procedure: Plotted histograms and boxplots of numerical features against fraud labels, and proportion bar charts of categorical features by fraud label.

Finding:

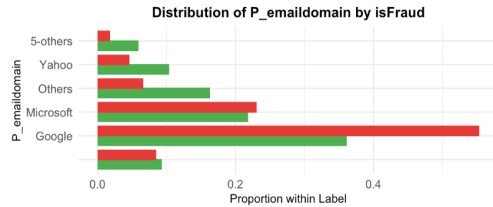


(Fig. 5: Bivariate histogram of C1 by Label)

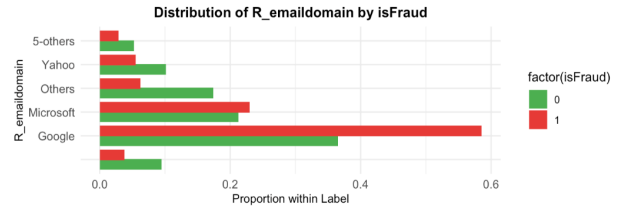


(Fig. 6: Bivariate histogram of C1 by Label)

Based on the bivariate histograms (Appx 8) and boxplots (Appx 9) of all numerical features, several variables exhibit different distributions across fraud labels, suggesting potential discriminatory power for fraud detection. Taking the feature “CI” as an example, figure 5 shows that fraud transactions are concentrated at relatively higher values compared with non-fraud transactions. The boxplot in figure 6 further confirms this pattern, with a higher median and wider interquartile range observed in the fraud class. This indicates that certain numerical features may capture abnormal transaction behaviour associated with fraud.



(Fig. 7: Bivariate bar of P_emaildomain by Label)



(Fig. 8: Bivariate bar of R_emaildomain by Label)

Furthermore, based on the bivariate bar charts (Appx 10) of categorical features, category proportions differ across fraud labels. Figures 7 and figure 8 show that “Google” email domains appear relatively more common in fraudulent transactions, while “Microsoft” domains are more frequently associated with non-fraud transactions. This suggests some categorical features contain useful signals for fraud detection.

Step 6 - Multivariate Analysis:

Procedure: Plotted the correlation table combining all the numerical features and the label.

Finding: Based on the heatmap (Appx 11), a representative section is presented in Figure 9. The results show that “isFraud” demonstrates relatively strong correlations with several numerical features, as indicated by higher color intensity along the label row, particularly “C1, C2, C7, C8, C10, C12, D8, id_01”. Also, the strong correlations among “TranID, TranDT, TranDTHour” suggest that these variables are highly related and may originate from the same underlying information source.



(Fig. 9: Correlation Heatmap of Numerical Features with Fraud Label)

Step 7: Feature engineering

Irrelevant Feature Removal: Remove the numerical features with an absolute correlation below 0.1 (Appx 11), since they show weak statistical association with fraudulent labels and are unlikely to contribute meaningful predictive information.

Categorical Feature Binning: As mentioned in Step 3, some categorical variables contain a large number of unique values. To reduce sparsity, each categorical feature was grouped into the top 20 most frequent categories, while the remaining values were combined into an “others” category.

Feature Creation & reasons: Based on the official dataset description (Appx 12), we have created 3 additional features to capture behavioural fraud patterns:

- 1) **“HourOfDay”:** derived from “TranDTHour % 24”, representing the transaction hour within a day. Fraudulent transactions may occur disproportionately during abnormal hours, whereas legitimate transactions are typically concentrated in daytime periods.
- 2) **“EmailMatch”:** derived from “P_emaildomain == R_emaildomain”. This feature is used to capture the relationship between purchaser and recipient ecosystems, which may differ between legitimate and fraudulent transaction flows.
- 3) **“IDMissingRatio”:** derived from the proportion of missing identity-related fields (Appx 12). It is likely that fraudulent users may intentionally provide incomplete identity information, leading to higher missing ratios compared with genuine customers.

Step 8: Discussion on the Key Findings

Finding 1: The dataset contains a noticeable proportion of missing information. In step 2, 10 features were identified as fully missing and 2 as constant-value features.

Insight & Hypothesis: These issues may result from data collection limitations or intentionally unrecorded information. Fraudulent users may avoid providing identifiable information. The transactions with higher missingness may be associated with fraud risk.

Finding 2: The dataset contains a noticeable proportion of outliers, with an average outlier rate of 11.18% (Step 4). In Step 4, we found transaction amounts are concentrated at lower values, while extreme values occur infrequently.

Insight & Hypothesis: The normal transactions may follow stable patterns, whereas fraudulent activities often deviate from typical behavior, such that extreme values may represent fraud signals rather than noise

Finding 3: The dataset is imbalanced as we found that the fraud label distribution shows that non-fraud transactions significantly exceed fraud cases in step 3.

Insight & Hypothesis: This distribution reflects real-world payment environments. Thus, we might need further data processing steps (e.g. resampling) to deal with this issue.

Finding 4: The counting features have a strong predictive power on fraud detection. In Step 6, the counting features demonstrated relatively higher correlations with the fraud label. In step 5, we found that fraud transactions tend to occur at higher C1 values compared with non-fraud cases.

Insight & Hypothesis: According to the official dataset description (**Appx 11**), C1 represents entity-related counts associated with payment cards or user activity. The legitimate users usually demonstrate consistent transaction behavior, leading to stable counts. On the other hand, Fraudsters may reuse cards, devices, or addresses across multiple suspicious attempts, increasing count-based metrics. As a result large values of counts might indicate abnormal usage patterns and may signal coordinated or repeated fraud attempts.

Finding 5: There are differences in fraud proportions across purchaser and recipient email domains. Both purchaser and recipient email domains suggest that certain providers (e.g. Google domains) appear more frequently in fraud transactions (Step 5).

Insight & Hypothesis: This may reflect differences in account creation convenience, anonymity, or usage patterns exploited by fraudsters.

Section 3: Fraud Modelling and Result Discussion

3.1 Modelling Strategy and Justification

This section was built on the findings reported in **Section 2**, considering the imbalanced class distribution and the presence of abnormal behaviour associated with fraudulent transactions. To ensure the reproducibility of the modelling process, a fixed random seed of 7410 was applied to all stochastic procedures in this section, such as the data partitioning and dataset rebalancing. The processed dataset was then partitioned into training and testing subsets using stratified random sampling with an 80:20 split where the fraud label distribution was preserved in both subsets. Given the strong class imbalance identified in the earlier analysis, SMOTE was applied to the training set only, while the testing set was kept unchanged to provide an unbiased out-of-sample evaluation.

We have developed two tree-based models, including a Decision Tree and a Random Forest. The Decision Tree was used as an interpretable baseline model since it can accommodate diverse data types, capturing nonlinear relationships, and producing decision rules that are easy to communicate in a business scenario. Also, we have pruned the Decision Tree after training for alleviating the issue of overfitting. On the other hand, the Random Forest was selected as the second model since it refines the decision tree through ensemble learning, thereby improving robustness and predictive performance. In the implementation, the Random Forest was trained using 100 trees, practically compromising between computational efficiency and model effectiveness. The use of these two models allowed us to compare an interpretable baseline against a stronger ensemble benchmark under the same fraud detection objective.

3.2 Performance Evaluation and Result Analysis

Model	Dataset	Accuracy	Recall	Precision	F1-Score	AUROC	AUPRC
Decision Tree	Train	0.876856	0.862744	0.874582	0.868623	0.910821	0.907244
	Test	0.860043	0.63544	0.421454	0.506784	0.808921	0.414798
Random Forest	Train	0.996486	0.992901	0.999651	0.996264	0.999981	0.999979
	Test	0.955498	0.711003	0.872087	0.78335	0.937859	0.841533

(Table 1: Evaluation result of the Decision Tree and the Random Forest)

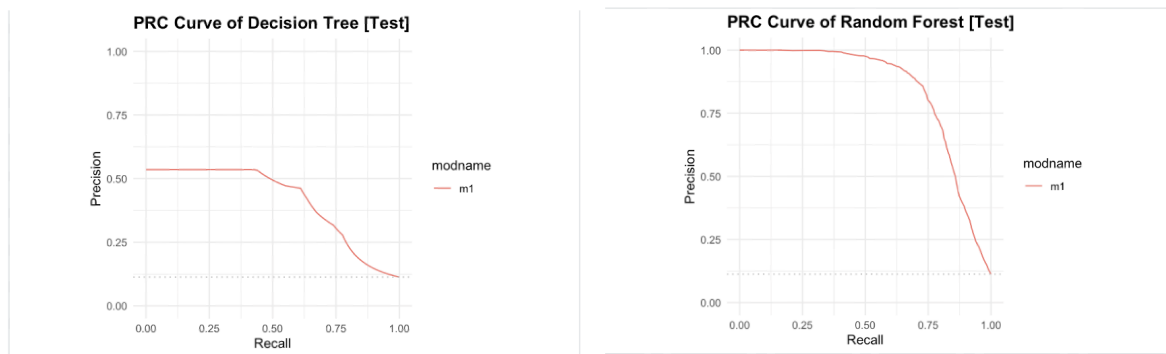
We have evaluated the performance of the Decision Tree and Random Forest models on both the training and testing datasets using six evaluation metrics, namely accuracy, recall, precision, F1-score, AUROC, and AUPRC. Based on an overview on **Table 1**, we can observe that both models show relatively high testing accuracy and AUROC values, while the other evaluation metrics are comparatively lower. As discussed in **Section 2**, the dataset is imbalanced, such that the accuracy and AUROC alone may overstate model performance as they are influenced by the majority non-fraud class. For that reason, this section will place more emphasis on recall, precision, F1-score, and AUPRC, aiming to better reflect the ability of the models to identify fraudulent transactions and control false alerts, so as to align with the objectives in **Section 1**.

Based on the testing results in **Table 1**, the Random Forest performed better than the Decision Tree across all six evaluation metrics. This suggests that the Random Forest was more effective in distinguishing fraudulent transactions from non-fraudulent ones. The improvement is particularly clear in recall and AUPRC, which are more important in fraud detection. However, when comparing the training and testing results, both models appear to show some degree of overfitting, since both models' recall and AUPRC fell from the training set to the testing set, indicating that they fitted the training data better than the unseen data.



(Fig. 10: Confusion matrices of the Decision Tree and Random Forest on the testing set)

Based on **Figure 10**, both models classified most non-fraud transactions correctly in the testing data, as shown by the relatively large number of correctly predicted non-fraud cases. However, the Random Forest produced a stronger classification result overall. Compared with the Decision Tree, it identified more fraudulent transactions correctly and generated fewer false fraud predictions. This suggests that the Random Forest achieved a better balance between fraud detection and false-positive control.

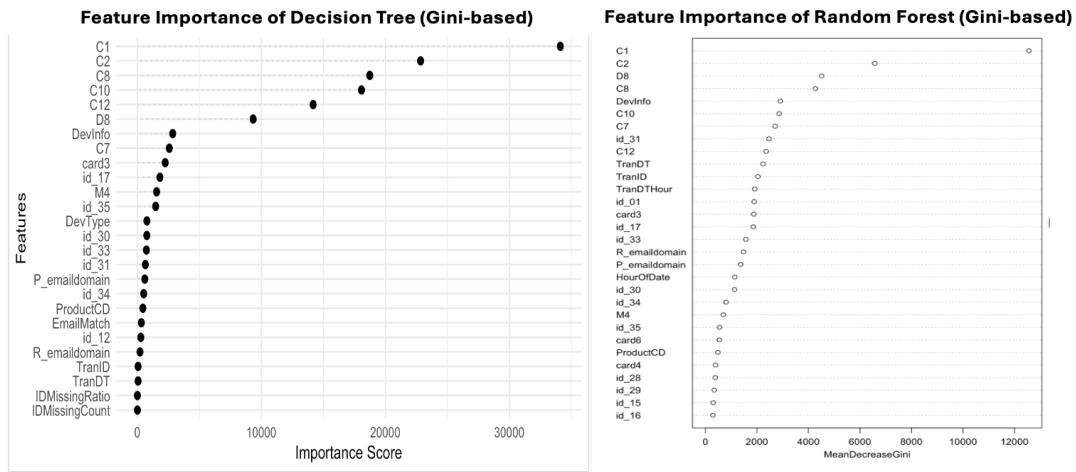


(Fig. 11: PRC Curves of the Decision Tree and Random Forest on the testing set)

Based on **Figure 11**, the Random Forest also shows a stronger Precision-Recall Curve than the Decision Tree. Its PRC is smoother and remains at a higher precision level over a wider range of recall values, whereas the PRC of the Decision Tree drops earlier and stays at a lower level. This indicates that the Random Forest was more stable in maintaining precision while increasing fraud capture. In other words, as the recall increased, the Random Forest was still able to keep precision relatively high, while the Decision Tree experienced a faster decline in precision. This pattern is consistent with the higher AUPRC value reported in **Table 1** and further supports that the Random Forest is the more effective model for this imbalanced fraud detection task.

To recap, the results from **Table 1**, **Figure 10**, and **Figure 11** also support the selection of the Random Forest as the final model for this case study. Compared with the Decision Tree, the Random Forest achieved stronger testing performance and showed a better balance between fraud detection and false-alert control. More importantly, it performed better in recall and AUPRC, which are significant for fraud detection in an imbalanced classification problem. This suggests that the ensemble nature of the Random Forest improved its performance. By combining multiple decision trees, the random forest model was more capable in producing robust predictions than a single Decision Tree and capturing the fraud patterns.

3.3 Important Features and Model Reasoning



(Fig. 12: : Variable Importance Comparison between Decision Tree and Random Forest Models (Gini-based))

Apart from the performance evaluation, we also conducted an analysis of feature importance. **Figure 12** shows that both the Decision Tree and Random Forest assign high importance to the counting features (e.g. C1, C2, C7, C8, C10, and C12), as well as the identity-related features (e.g. DevInfo and id_31). This suggests that both models generally agree on the main predictors of fraudulent transactions and that these variables contribute consistently to the classification process.

These findings are consistent with **Section 2 Key Finding 4**, which showed that the counting features had relatively stronger relationships with the fraud label. In the earlier bivariate analysis, fraudulent transactions were found to occur at higher C1 values than non-fraud transactions. Therefore, the feature-importance results in **Figure 12** provide further support for the view that the counting variables contain strong discriminatory power for fraud detection. A possible explanation is that these counting features capture repeated or concentrated activity associated with the same card, user, device, or transaction environment. Legitimate users usually show more regular and stable transaction behaviour, whereas fraudulent activities may involve repeated attempts, abnormal usage intensity, or coordinated reuse of related entities. As a result, unusually large values in these counting variables may indicate suspicious transaction patterns and help the models distinguish fraudulent cases from genuine ones.

Furthermore, the high importance of DevInfo and id_31 suggests that digital identity characteristics provide useful additional information for fraud detection. Fraudulent transactions may be associated with particular browser types, device environments, or technical configurations that differ from those of legitimate users. This implies that, beyond transaction counts alone, device and identity signals also play a meaningful role in capturing fraud-related behaviour.

3.4 Strengths, Limitations and Improvement Directions

To recap, both tree-based models can capture nonlinear relationships, while the Decision Tree is easier to interpret through its splitting structure (**Appendix 13**). Compared with the Decision Tree, the Random Forest showed better generalisation on unseen data and a better balance between fraud detection and false-positive control, since its final prediction is based on the combined voting of many decision trees, which helps improve predictive robustness and reduce the instability of a single-tree model. As a result, the Random Forest is more suitable for capturing the complex relationships among the features of fraudulent transactions.

Nonetheless, tree-based models still have several limitations. First, the gap between training and testing performance suggests that some overfitting may still exist. Second, the current analysis mainly relied on basic model settings, so further hyperparameter tuning may improve performance. Third, the Random Forest requires more runtime than a single Decision Tree because it aggregates the results of many trees, which increases computational cost.

Based on these findings, several improvements are recommended. Future work may conduct hyperparameter tuning to optimise model settings, such as the number of trees, the number of variables considered at each split, and other tree-growth controls, so as to improve predictive performance and reduce the risk of overfitting. It may also adjust the classification threshold according to the institution's fraud-control objective, since different thresholds imply different trade-offs between fraud detection and false-positive control, particularly in an imbalanced fraud setting where precision, recall, and the Precision-Recall curve are important evaluation criteria.

Section 4: Non-data Analytics for Transaction Fraud

4.1 Risks and Red Flags

In addition to the quantitative modeling results, we have also analysed the non-data analytic elements that drive transaction risk. The impact of transaction fraud creates four major types of risk, namely the financial, operational, reputational, and regulatory risk. Financial risk involves the direct loss of money from unauthorized payments and the extra cost of chargeback fees. These losses are made worse by operational strain, since the upsurge in fraud cases requires resources for manual checks and legal follow-up. Furthermore, frequent security issues would damage a company's reputation, causing customers to lose trust and stop using the service. Finally, these weaknesses lead to regulatory risk, where failing to meet proper detection and reporting standards can result in breaking anti-money laundering laws and facing heavy government fines.

To mitigate transaction risks, we have to monitor behavioral patterns deviating from legitimate user norms. High-risk indicators often manifest as velocity anomalies, where a sudden increase in transaction frequency suggests automated attacks or unauthorized card testing. Furthermore, technical red flags, such as rapid successions of IP address changes or multiple device handoffs, frequently signal account takeovers by external actors. In addition, temporal anomalies involving large cross-border transactions during unusual hours can highlight geographic discrepancies and unauthorized access from different time zones. These non-data analytic signals align with the findings of this study, as they emphasize how tracking inconsistent digital footprints is essential for preventing transaction fraud.

4.2 Governance and Control Measures

The red flags above should not only be viewed as abnormal data patterns, but also as indicators of potential weaknesses in onboarding, authentication, approval, and monitoring processes. From a control perspective, an important objective is to reduce the opportunity element in the fraud triangle by making it more difficult for suspicious actors to exploit weak procedures or override existing controls. This requires a strong governance framework, including a clear anti-fraud policy, a code of conduct, fraud-awareness training, effective management oversight, and a consistent escalation process for suspicious activities.

In operational terms, institutions should apply a risk-based approach to customer due diligence and ongoing monitoring. This requires a due diligence process that covers the full customer lifecycle, from initial identity verification to periodic review of account activity. By properly documenting the context of suspicious transactions and maintaining a clear audit trail, institutions can ensure that high-risk activities remain transparent and traceable.

In addition, internal controls should be reinforced through independent oversight, clear accountability, and effective authorization procedures. Confidential disclosure channels can further strengthen the control environment by supporting fraud detection, human supervision, and internal audit functions.

4.3 Suggestion: Three-Layer Prevention Plan

Based on the analytical findings, a three-layer defense strategy is recommended to enhance future transaction fraud management. The first layer focuses on immediate prevention by implementing rigorous controls at the point of entry. Rather than allowing all transactions to reach the modeling stage, this layer uses foundational authentication and access barriers to filter out obvious high-risk patterns during onboarding and initiation. This proactive approach reduces the volume of fraudulent attempts before they can enter the payment ecosystem.

The second layer integrates the advanced machine learning capabilities developed in this study. By utilizing the outstanding predictive power of the Random Forest model, institutions can move beyond simple rules to identify complex fraud patterns through sophisticated risk scoring. This analytical layer allows for the prioritization of suspicious activities, ensuring that the most critical threats are identified with higher accuracy than traditional methods.

The third layer provides the essential human oversight needed to validate and act upon model outputs. This involves a structured investigation process where specialized staff review flagged alerts and assess broader control weaknesses to prevent future reoccurrence. Furthermore, institutional effectiveness is strengthened through continuous internal audits and the sharing of threat intelligence across the industry. This integrated approach is far more effective than any standalone tool since it combines data-driven insights with robust governance and professional accountability.

Section 5: Conclusion

This case study demonstrates that effective fraud management requires a synergy between advanced machine learning and robust organizational controls. The comparative analysis reveals that the Random Forest model provides superior predictive performance and stability compared to the Decision Tree particularly when identifying complex behavioral patterns. By accurately capturing high risk indicators, such as velocity anomalies and inconsistent digital footprints, this data driven approach establishes a reliable foundation for real time detection. To further refine these capabilities, future work should focus on hyperparameter tuning and threshold optimization to align model sensitivity with specific institutional fraud control objectives. Nevertheless, the findings also suggest that technical tools are most valuable when they are embedded within a wider framework of governance, control, and human oversight.

The primary recommendation for future transaction fraud management is the adoption of the three layer prevention plan. This strategy ensures that automated insights from the Random Forest model are supported by immediate rule based filtering and specialized human investigation. Strengthening corporate governance through clear anti-fraud policies and the segregation of duties further ensures that these systems operate within a transparent and accountable environment. Ultimately this integrated approach allows institutions to mitigate financial and regulatory risks while maintaining a seamless experience for legitimate users. By combining analytical precision with strong internal controls financial institutions can build a proactive defense that is capable of adapting to the evolving landscape of digital financial crime in the future.

Appendix

Appendix 1: Structure of Dataset

'data.frame': 100000 obs. of 101 variables:

\$ TranID	: int	2987004 2987011 2987016 2987022 2987038	\$ M1	: logi	NA NA NA NA NA NA NA ...
2987040 2987048 2987057 2987069 2987070 ...			\$ M2	: logi	NA NA NA NA NA NA NA ...
\$ isFraud	: int	0 0 0 0 0 0 0 0 0 ...	\$ M3	: logi	NA NA NA NA NA NA NA ...
\$ TranDT	: int	86506 86555 86620 86786 87172 87209	\$ M4	: chr	"" "M0" "" "" ...
87317 87445 87725 87735 ...			\$ M5	: logi	NA NA NA NA NA NA NA ...
\$ TranDTHour	: int	24 24 24 24 24 24 24 24 24 ...	\$ M6	: logi	NA NA NA NA NA NA NA ...
\$ TranAmt	: num	69.3 23.1 41.8 69.3 34.9 ...	\$ M7	: logi	NA NA NA NA NA NA NA ...
\$ ProductCD	: chr	"TP" "LY" "TP" "TP" ...	\$ M8	: logi	NA NA NA NA NA NA NA ...
\$ addr1	: int	402 NA 152 281 312 NA NA 186 312 307 ...	\$ M9	: logi	NA NA NA NA NA NA NA ...
\$ addr2	: int	73 NA 73 73 73 NA NA 73 73 73 ...	\$ V310	: num	0 0 0 0 145 0 0 0 0 ...
\$ dist1	: logi	NA NA NA NA NA NA NA ...	\$ V311	: num	0 0 0 0 0 0 0 0 0 0 ...
\$ dist2	: int	NA 30 NA NA NA 98 NA NA NA 84 NA ...	\$ V312	: num	0 0 0 0 25 0 0 0 0 ...
\$ card1	: int	4497 4461 1790 1724 5463 13329 15885 11839	\$ V313	: num	0 0 0 0 0 0 0 0 0 0 ...
12866 3682 ...			\$ V314	: num	0 0 0 0 0 0 0 0 0 0 ...
\$ card2	: int	514 375 555 583 399 569 545 490 303 264 ...	\$ id_01	: int	0 -5 0 -15 0 -10 -5 0 0 ...
\$ card3	: int	150 185 150 150 150 117 185 150 150 150 ...	\$ id_02	: int	70787 221832 7460 NA 31964 116098 257037
\$ card4	: chr	"mastercard" "mastercard" "visa" "visa" ...	88525 69542 132356 ...		
\$ card5	: int	102 224 226 226 137 226 138 226 226 162 ...	\$ id_03	: int	NA NA 0 NA 0 0 NA NA 0 NA ...
\$ card6	: chr	"credit" "debit" "debit" "credit" ...	\$ id_04	: int	NA NA 0 NA 0 0 NA NA 0 NA ...
\$ P_emaildomain	: chr	"Google" "Microsoft" "AOL" "Google" ...	\$ id_05	: int	NA 0 1 NA 0 0 0 NA 2 1 ...
\$ R_emaildomain	: chr	"" "Microsoft" "" "" ...	\$ id_06	: int	NA -6 0 NA -10 0 0 NA -4 -6 ...
\$ C1	: int	1 1 1 1 1 2 1 1 7 1 ...	\$ id_07	: int	NA NA NA NA NA NA NA NA NA NA ...
\$ C2	: int	1 1 1 1 2 5 1 1 1 1 ...	\$ id_08	: int	NA NA NA NA NA NA NA NA NA NA ...
\$ C3	: int	0 0 0 0 0 0 0 0 0 0 ...	\$ id_09	: int	NA NA 0 NA 0 0 NA NA 0 NA ...
\$ C4	: int	0 1 0 0 1 1 1 0 3 0 ...	\$ id_10	: int	NA NA 0 NA 0 0 NA NA 0 NA ...
\$ C5	: int	0 0 0 0 0 0 0 0 0 0 ...	\$ id_11	: num	100 100 100 NA 100 100 100 100 100 100 ...
\$ C6	: int	1 1 1 1 0 1 1 1 0 1 ...	\$ id_12	: chr	"NotFound" "NotFound" "NotFound"
\$ C7	: int	0 1 0 0 0 1 1 0 0 0 ...	"NotFound" ...		
\$ C8	: int	1 1 1 0 6 1 1 1 5 1 ...	\$ id_13	: int	NA 52 NA 14 NA 52 52 NA NA NA ...
\$ C9	: int	0 0 0 0 0 0 0 0 0 0 ...	\$ id_14	: int	-480 NA -300 NA -300 NA NA -300 -300 -
\$ C10	: int	1 1 1 0 93 1 1 1 1 1 1 ...	300 ...		
\$ C11	: int	1 1 1 1 1 2 1 1 1 1 ...	\$ id_15	: chr	"New" "New" "Found" "" ...
\$ C12	: int	0 1 0 0 0 2 1 0 0 0 ...	\$ id_16	: chr	"NotFound" "NotFound" "Found" "" ...
\$ C13	: int	1 1 1 1 96 3 1 1 12 1 ...	\$ id_17	: int	166 225 166 NA 166 121 225 166 166 166 ...
\$ C14	: int	1 1 1 1 1 2 1 1 6 1 ...	\$ id_18	: int	NA NA 15 NA 15 NA NA NA 15 13 ...
\$ D1	: int	0 0 0 0 542 0 0 0 0 0 ...	\$ id_19	: int	542 176 529 NA 352 410 484 278 352 529 ...
\$ D2	: int	NA NA NA NA NA 542 NA NA NA NA NA ...	\$ id_20	: int	144 507 575 NA 533 142 507 600 533 549 ...
\$ D3	: int	NA NA NA NA NA NA NA NA NA NA NA ...	\$ id_21	: int	NA NA NA NA NA NA NA NA NA NA NA ...
\$ D4	: int	NA 0 NA NA NA NA 0 0 0 NA NA ...	\$ id_22	: int	NA NA NA NA NA NA NA NA NA NA NA ...
\$ D5	: int	NA NA NA NA NA NA 0 NA NA NA NA ...	\$ id_23	: chr	"" "" "" "" "" ...
\$ D6	: int	NA 0 NA NA NA 537 0 0 NA 216 NA ...	\$ id_24	: int	NA NA NA NA NA NA NA NA NA NA NA ...
\$ D7	: int	NA NA NA NA NA 4 0 NA NA 4 NA ...	\$ id_25	: int	NA NA NA NA NA NA NA NA NA NA NA ...
\$ D8	: num	NA NA 26 NA 29 83 NA NA 126 NA ...	\$ id_26	: int	NA NA NA NA NA NA NA NA NA NA NA ...
\$ D9	: num	NA NA 0 NA 0 0 NA NA 0 NA ...	\$ id_27	: chr	"" "" "" "" "" ...
\$ D10	: int	NA 0 NA NA NA 520 0 0 NA 284 NA ...	\$ id_28	: chr	"New" "New" "Found" "" ...
\$ D11	: logi	NA NA NA NA NA NA NA ...	\$ id_29	: chr	"NotFound" "NotFound" "Found" "" ...
\$ D12	: int	NA 0 NA NA NA NA 0 0 NA NA NA ...	\$ id_30	: chr	"Android" "" "Mac" "" ...
\$ D13	: int	NA 0 NA NA NA NA 0 0 NA NA NA ...	\$ id_31	: chr	"samsung browser 6.2" "chrome 62.0" "chrome
\$ D14	: int	NA 0 NA NA NA NA 0 0 NA NA NA ...	62.0" "" ...		
\$ D15	: int	NA 0 NA NA NA 541 0 0 0 284 NA ...	\$ id_32	: int	32 NA 24 NA 32 NA NA 32 32 32 ...
			\$ id_33	: chr	"2220x1080" "" "1280x800" "" ...
			\$ id_34	: chr	"match_status:2" "" "match_status:2" "" ...
			\$ id_35	: logi	TRUE FALSE TRUE NA TRUE FALSE ...
			\$ id_36	: logi	FALSE FALSE FALSE NA FALSE FALSE ...
			\$ id_37	: logi	TRUE TRUE TRUE NA TRUE TRUE ...
			\$ id_38	: logi	TRUE TRUE TRUE NA TRUE TRUE ...

Appendix 2: Summary of Dataset

TranID	isFraud	TranDT	TranDTHour	TranAmt
Min. : 2987004	Min. : 0.0000	Min. : 86506	Min. : 24	Min. : 0.7918
1st Qu.: 3078716	1st Qu.: 0.0000	1st Qu.: 1900286	1st Qu.: 527	1st Qu.: 35.4393
Median : 3203512	Median : 0.0000	Median : 5031202	Median : 1397	Median : 69.3070
Mean : 3238037	Mean : 0.1132	Mean : 6209720	Mean : 1724	Mean : 115.6799
3rd Qu.: 3393812	3rd Qu.: 0.0000	3rd Qu.: 10272711	3rd Qu.: 2853	3rd Qu.: 138.1970
Max. : 3577534	Max. : 1.0000	Max. : 15811030	Max. : 4391	Max. : 2480.4570
ProductCD	addr1	addr2	dist1	dist2
Length:100000	Min. : 82.0	Min. : 18.00	Mode:logical	Min. : 0.0
Class :character	1st Qu.:186.0	1st Qu.: 73.00	NA's:100000	1st Qu.: 7.0
Mode :character	Median :281.0	Median : 73.00		Median : 38.0
	Mean :278.6	Mean : 73.01		Mean : 230.5
	3rd Qu.:312.0	3rd Qu.: 73.00		3rd Qu.: 207.0
	Max. :522.0	Max. :110.00		Max. :11623.0
	NA's :42881	NA's :42881		NA's :73694
card1	card2	card3	card4	card5
Min. : 1000	Min. :100.0	Min. :100.0	Length:100000	Min. :100.0
1st Qu.: 5666	1st Qu.:264.0	1st Qu.:150.0	Class :character	1st Qu.:138.0
Median : 9633	Median :408.0	Median :150.0	Mode :character	Median :224.0
Mean : 9849	Mean :389.7	Mean :162.1		Mean :190.7
3rd Qu.:15063	3rd Qu.:545.0	3rd Qu.:185.0		3rd Qu.:226.0
Max. :18396	Max. :600.0	Max. :231.0		Max. :237.0
	NA's :688	NA's :115		NA's :660
card6	P_emaildomain	R_emaildomain	C1	C2
Length:100000	Length:100000	Length:100000	Min. : 0.00	Min. : 0.00
Class :character	Class :character	Class :character	1st Qu.: 1.00	1st Qu.: 1.00
Mode :character	Mode :character	Mode :character	Median : 1.00	Median : 1.00
			Mean : 29.12	Mean : 36.08
			3rd Qu.: 2.00	3rd Qu.: 2.00
			Max. :4684.00	Max. :5691.00
C3	C4	C5	C6	C7
Min. : 0.00000	Min. : 0.00	Min. : 0	Min. : 0.00	Min. : 0.00
1st Qu.: 0.00000	1st Qu.: 1.00	1st Qu.: 0	1st Qu.: 1.00	1st Qu.: 0.00
Median : 0.00000	Median : 1.00	Median : 0	Median : 1.00	Median : 0.00
Mean : 0.02083	Mean : 16.15	Mean : 0	Mean : 16.13	Mean : 10.93
3rd Qu.: 0.00000	3rd Qu.: 1.00	3rd Qu.: 0	3rd Qu.: 1.00	3rd Qu.: 1.00
Max. :23.00000	Max. :2252.00	Max. : 0	Max. :2252.00	Max. :2254.00
C8	C9	C10	C11	C12
Min. : 0.00	Min. : 0	Min. : 0.00	Min. : 0.0	Min. : 0.0
1st Qu.: 1.00	1st Qu.: 0	1st Qu.: 1.00	1st Qu.: 1.0	1st Qu.: 0.0
Median : 1.00	Median : 0	Median : 1.00	Median : 1.0	Median : 0.0
Mean : 20.17	Mean : 0	Mean : 20.42	Mean : 21.2	Mean : 15.5
3rd Qu.: 2.00	3rd Qu.: 0	3rd Qu.: 2.00	3rd Qu.: 1.0	3rd Qu.: 1.0
Max. :3330.00	Max. : 0	Max. :3256.00	Max. :3187.0	Max. :3187.0
C13	C14	D1	D2	D3
Min. : 0.00	Min. : 0.00	Min. : 0.00	Min. : 0.0	Min. : 0.00
1st Qu.: 1.00	1st Qu.: 1.00	1st Qu.: 0.00	1st Qu.: 2.0	1st Qu.: 0.00
Median : 1.00	Median : 1.00	Median : 0.00	Median : 36.0	Median : 1.00
Mean : 20.93	Mean : 10.77	Mean : 28.83	Mean :121.3	Mean : 34.78
3rd Qu.: 2.00	3rd Qu.: 1.00	3rd Qu.: 0.00	3rd Qu.:203.0	3rd Qu.: 16.00
Max. :2917.00	Max. :1428.00	Max. :640.00	Max. :640.0	Max. :819.00
		NA's :155	NA's :78733	NA's :79692
D4	D5	D6	D7	D8
Min. : -74.00	Min. : 0.0	Min. : -74	Min. : 0.00	Min. : 0.000
1st Qu.: 0.00	1st Qu.: 0.0	1st Qu.: 0	1st Qu.: 0.00	1st Qu.: 0.958
Median : 0.00	Median : 0.0	Median : 0	Median : 0.00	Median : 33.708
Mean : 59.89	Mean : 43.8	Mean : 69	Mean : 39.23	Mean : 141.362
3rd Qu.: 16.00	3rd Qu.: 17.0	3rd Qu.: 39	3rd Qu.: 14.00	3rd Qu.: 178.750
Max. :864.00	Max. :775.0	Max. :873	Max. :843.00	Max. :1517.833
NA's :54271	NA's :76124	NA's :52588	NA's :74148	NA's :47606
D9	D10	D11	D12	D13
Min. :0.000	Min. : 0.00	Mode:logical	Min. : -74.00	Min. : 0.00
1st Qu.:0.208	1st Qu.: 0.00	NA's:100000	1st Qu.: 0.00	1st Qu.: 0.00
Median :0.667	Median : 0.00		Median : 0.00	Median : 0.00
Mean :0.558	Mean : 21.68		Mean : 52.88	Mean : 17.26
3rd Qu.:0.833	3rd Qu.: 0.00		3rd Qu.: 13.00	3rd Qu.: 0.00
Max. :0.958	Max. :866.00		Max. :648.00	Max. :847.00
NA's :47606	NA's :51253		NA's :58199	NA's :56376
D14	D15	M1	M2	M3
Min. : -193.0	Min. : -74.00	Mode:logical	Mode:logical	Mode:logical
1st Qu.: 0.0	1st Qu.: 0.00	NA's:100000	NA's:100000	NA's:100000
Median : 0.0	Median : 0.00			
Mean : 57.5	Mean : 52.79			
3rd Qu.: 2.0	3rd Qu.: 5.00			
Max. : 878.0	Max. :878.00			
NA's :56054	NA's :51915			
M4	M5	M6	M7	M8
Length:100000	Mode:logical	Mode:logical	Mode:logical	Mode:logical
Class :character	NA's:100000	NA's:100000	NA's:100000	NA's:100000
Mode :character				

M9	V310	V311	V312	V313
Mode:logical	Min. : 0.00	Min. : 0.000	Min. : 0.00	Min. : 0.000
NA's:100000	1st Qu.: 0.00	1st Qu.: 0.000	1st Qu.: 0.00	1st Qu.: 0.000
	Median : 0.00	Median : 0.000	Median : 0.00	Median : 0.000
	Mean : 35.91	Mean : 4.707	Mean : 18.04	Mean : 7.972
	3rd Qu.: 0.00	3rd Qu.: 0.000	3rd Qu.: 0.00	3rd Qu.: 0.000
	Max. :55125.00	Max. :55125.000	Max. :55125.00	Max. :2000.000
	NA's :1	NA's :1	NA's :1	NA's :155
V314	id_01	id_02	id_03	id_04
Min. : 0.00	Min. : -100.00	Min. : 1	Min. : -13.000	Min. : -28.000
1st Qu.: 0.00	1st Qu.: -10.00	1st Qu.: 68568	1st Qu.: 0.000	1st Qu.: 0.000
Median : 0.00	Median : -5.00	Median :126869	Median : 0.000	Median : 0.000
Mean : 15.99	Mean : -10.41	Mean :175813	Mean : 0.064	Mean : -0.066
3rd Qu.: 0.00	3rd Qu.: -5.00	3rd Qu.:229899	3rd Qu.: 0.000	3rd Qu.: 0.000
Max. :2250.00	Max. : 0.00	Max. :999595	Max. : 10.000	Max. : 0.000
NA's :155		NA's :2276	NA's :53566	NA's :53566
id_05	id_06	id_07	id_08	id_09
Min. : -72.000	Min. : -100.00	Min. : -46.00	Min. : -100.00	Min. : -36.000
1st Qu.: 0.000	1st Qu.: -6.00	1st Qu.: 4.00	1st Qu.: -48.00	1st Qu.: -360.0
Median : 0.000	Median : 0.00	Median : 14.00	Median : -34.00	Median : 0.000
Mean : 1.623	Mean : -6.82	Mean : 13.25	Mean : -38.76	Mean : 0.096
3rd Qu.: 1.000	3rd Qu.: 0.00	3rd Qu.: 22.00	3rd Qu.: -23.00	3rd Qu.: 0.000
Max. : 52.000	Max. : 0.00	Max. : 61.00	Max. : 0.00	Max. : 25.000
NA's :5033	NA's :5033	NA's :96414	NA's :96414	NA's :47606
id_10	id_11	id_12	id_13	id_14
Min. : -100.0	Min. : 90.00	Length:100000	Min. :10.00	Min. : -660.0
1st Qu.: 0.0	1st Qu.:100.00	Class :character	1st Qu.:49.00	1st Qu.: -360.0
Median : 0.0	Median :100.00	Mode :character	Median :52.00	Median : -300.0
Mean : -0.3	Mean : 99.75		Mean :48.03	Mean : -343.8
3rd Qu.: 0.0	3rd Qu.:100.00		3rd Qu.:52.00	3rd Qu.: -300.0
Max. : 0.0	Max. :100.00		Max. :64.00	Max. : 720.0
NA's :47606	NA's :2203		NA's :11528	NA's :45478
id_15	id_16	id_17	id_18	id_19
Length:100000	Length:100000	Min. :100.0	Min. :10.00	Min. :100.0
Class :character	Class :character	1st Qu.:166.0	1st Qu.:13.00	1st Qu.:266.0
Mode :character	Mode :character	Median :166.0	Median :15.00	Median :339.0
		Mean :190.1	Mean :14.24	Mean :352.4
		3rd Qu.:225.0	3rd Qu.:15.00	3rd Qu.:427.0
		Max. :229.0	Max. :29.00	Max. :671.0
		NA's :3337	NA's :68424	NA's :3374
id_20	id_21	id_22	id_23	id_24
Min. :100.0	Min. :100.0	Min. :12.0	Length:100000	Min. :11.00
1st Qu.:266.0	1st Qu.:252.0	1st Qu.:14.0	Class :character	1st Qu.:11.00
Median :472.0	Median :252.0	Median :14.0	Mode :character	Median :11.00
Mean :405.2	Mean :369.1	Mean :16.1		Mean :12.79
3rd Qu.:533.0	3rd Qu.:489.5	3rd Qu.:14.0		3rd Qu.:15.00
Max. :660.0	Max. :854.0	Max. :44.0		Max. :26.00
NA's :3415	NA's :96409	NA's :96405		NA's :96683
id_25	id_26	id_27	id_28	id_29
Min. :100.0	Min. :100.0	Length:100000	Length:100000	Length:100000
1st Qu.:321.0	1st Qu.:119.0	Class :character	Class :character	Class :character
Median :321.0	Median :149.0	Mode :character	Mode :character	Mode :character
Mean :329.6	Mean :149.6			
3rd Qu.:371.0	3rd Qu.:169.0			
Max. :548.0	Max. :216.0			
NA's :96434	NA's :96410			
id_30	id_31	id_32	id_33	id_34
Length:100000	Length:100000	Min. : 0.00	Length:100000	Length:100000
Class :character	Class :character	1st Qu.:24.00	Class :character	Class :character
Mode :character	Mode :character	Median :24.00	Mode :character	Mode :character
		Mean :26.52		
		3rd Qu.:32.00		
		Max. :32.00		
		NA's :47177		
id_35	id_36	id_37	id_38	DevType
Mode :logical	Mode :logical	Mode :logical	Mode :logical	Length:100000
FALSE:44814	FALSE:93074	FALSE:20964	FALSE:51740	Class :character
TRUE :52990	TRUE :4730	TRUE :76840	TRUE :46064	Mode :character
NA's :2196	NA's :2196	NA's :2196	NA's :2196	

```

DevInfo
Length:100000
Class :character
Mode :character

```

Appendix 3: Missing rate of each features

TranID	isFraud	TranDT	TranDTHour	TranAmt	ProductCD
0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
addr1	addr2	dist1	dist2	card1	card2
0.42881	0.42881	1.00000	0.73694	0.00000	0.00688
card3	card4	card5	card6	P_emaildomain	R_emaildomain
0.00115	0.00000	0.00660	0.00000	0.00000	0.00000
C1	C2	C3	C4	C5	C6
0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
C7	C8	C9	C10	C11	C12
0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
C13	C14	D1	D2	D3	D4
0.00000	0.00000	0.00155	0.78733	0.79692	0.54271
D5	D6	D7	D8	D9	D10
0.76124	0.52588	0.74148	0.47606	0.47606	0.51253
D11	D12	D13	D14	D15	M1
1.00000	0.58199	0.56376	0.56054	0.51915	1.00000
M2	M3	M4	M5	M6	M7
1.00000	1.00000	0.00000	1.00000	1.00000	1.00000
M8	M9	V310	V311	V312	V313
1.00000	1.00000	0.00001	0.00001	0.00001	0.00155
V314	id_01	id_02	id_03	id_04	id_05
0.00155	0.00000	0.02276	0.53566	0.53566	0.05033
id_06	id_07	id_08	id_09	id_10	id_11
0.05033	0.96414	0.96414	0.47606	0.47606	0.02203
id_12	id_13	id_14	id_15	id_16	id_17
0.00000	0.11528	0.45478	0.00000	0.00000	0.03337
id_18	id_19	id_20	id_21	id_22	id_23
0.68424	0.03374	0.03415	0.96409	0.96405	0.00000
id_24	id_25	id_26	id_27	id_28	id_29
0.96683	0.96434	0.96410	0.00000	0.00000	0.00000
id_30	id_31	id_32	id_33	id_34	id_35
0.00000	0.00000	0.47177	0.00000	0.00000	0.02196
id_36	id_37	id_38	DevType	DevInfo	
0.02196	0.02196	0.02196	0.00000	0.00000	

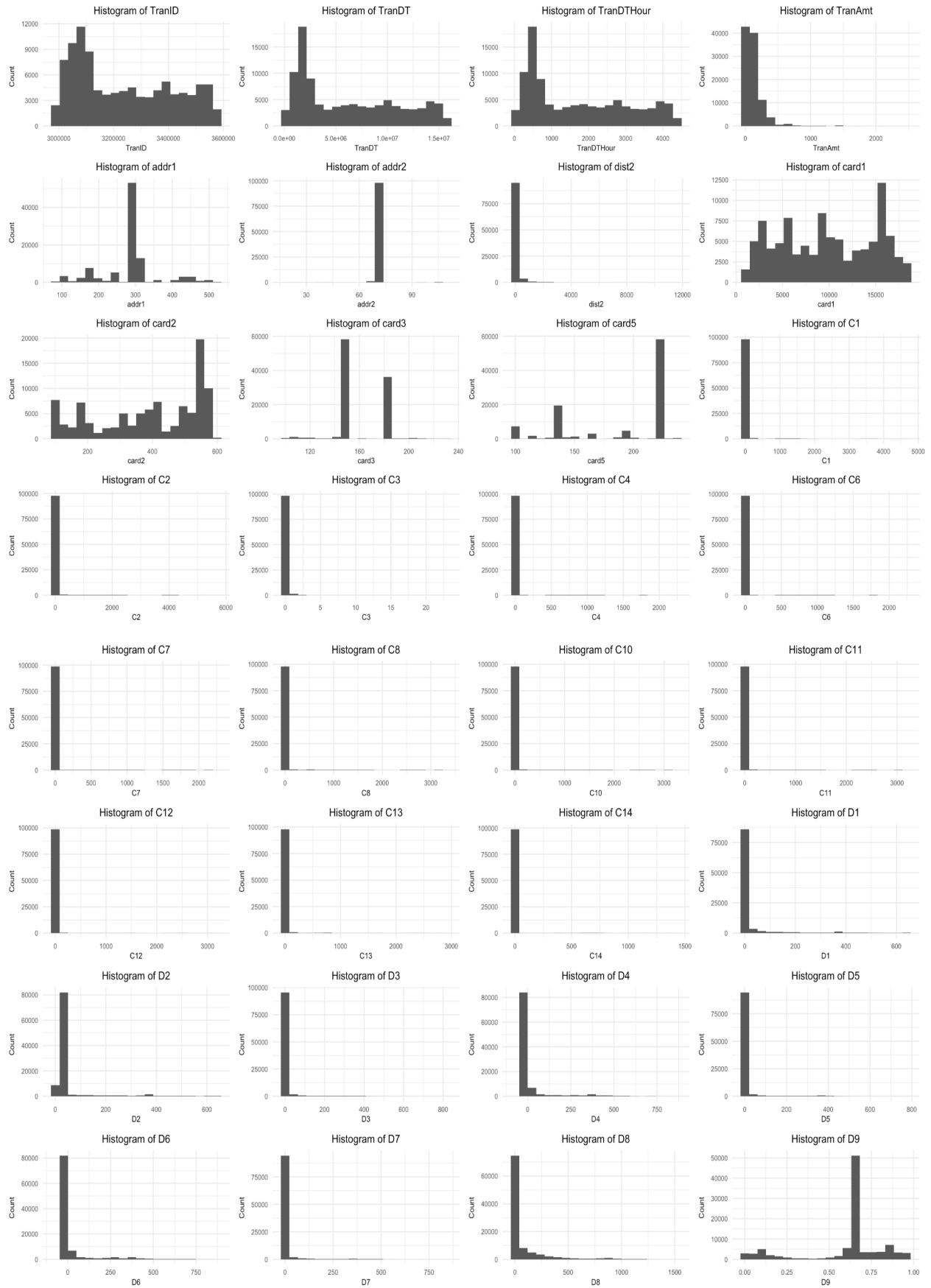
Features with 100% missing value [Removed]

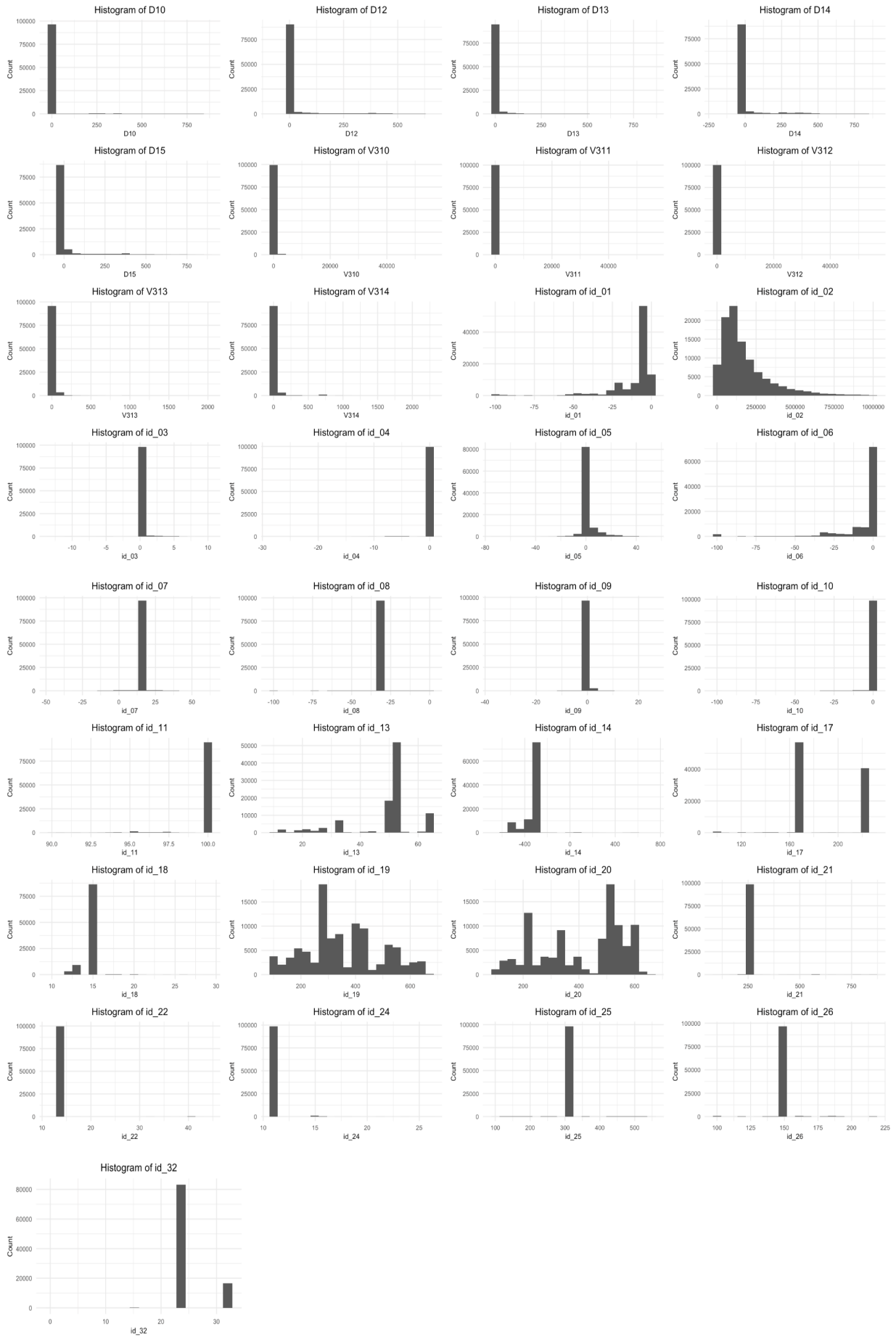
"dist1" "D11" "M1" "M2" "M3" "M5" "M6" "M7" "M8" "M9"

Features with constant value [Removed]

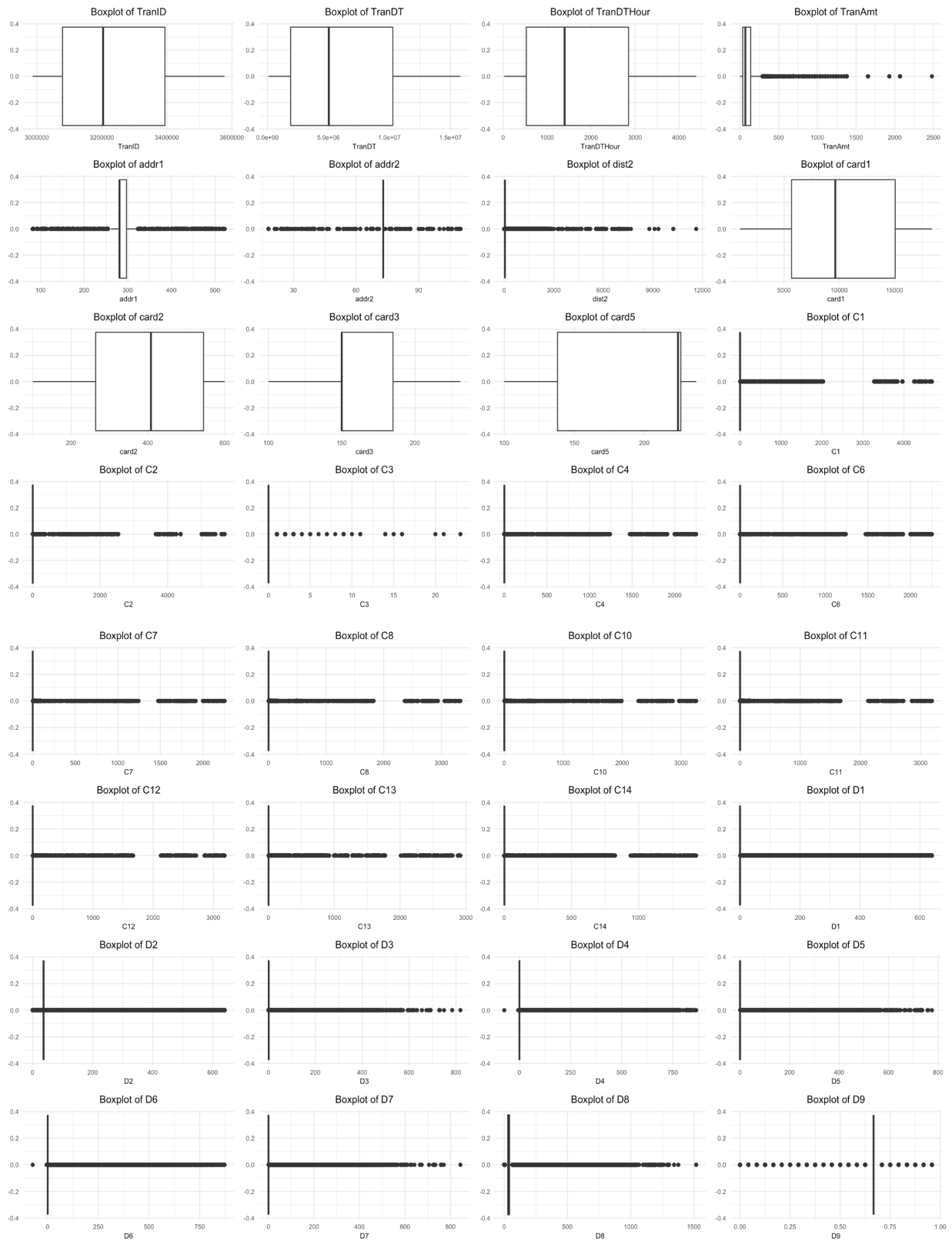
"C5" "C9"

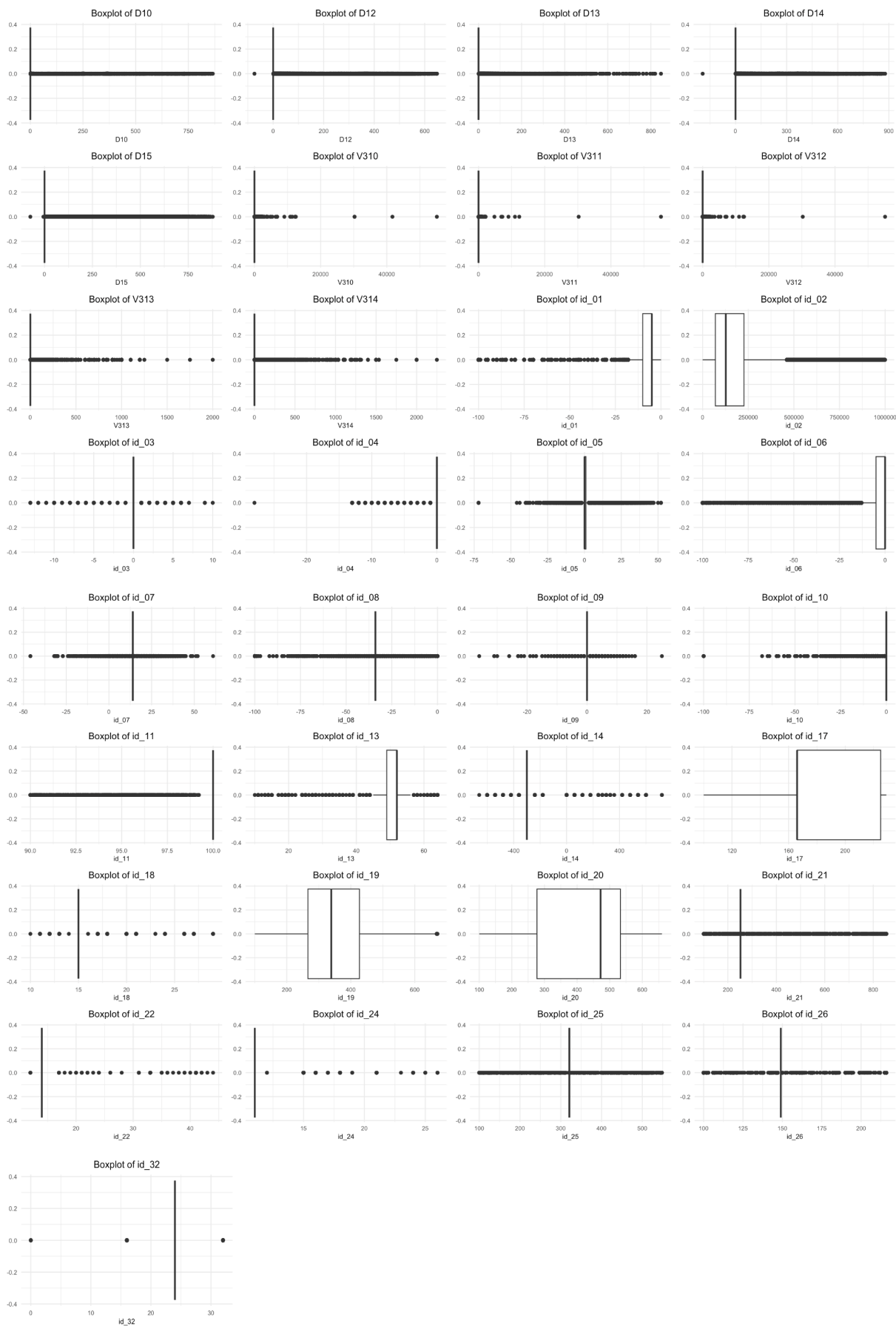
Appendix 4: Histogram of every numerical feature



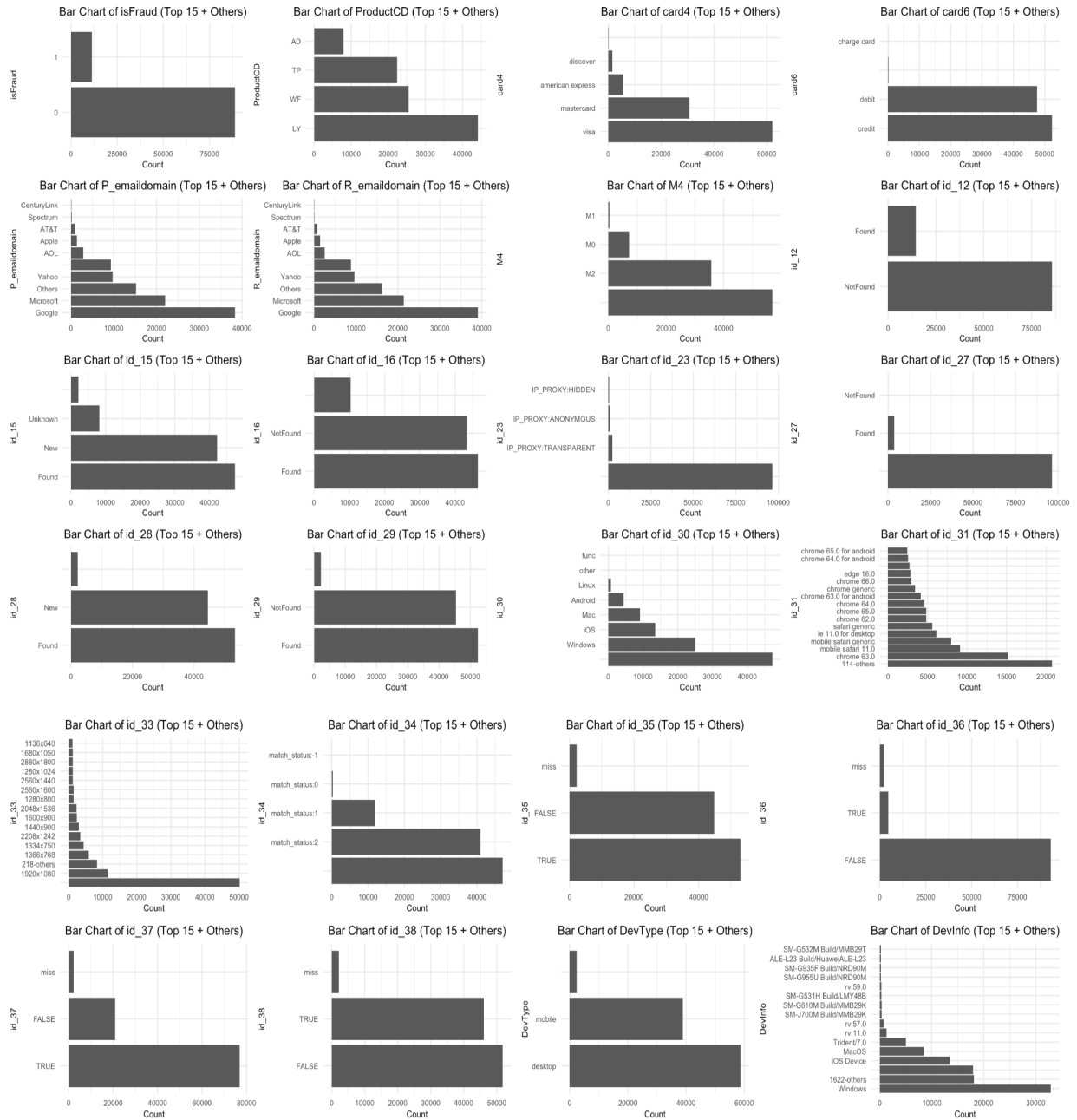


Appendix 5: Boxplot of every numerical feature





Appendix 6: Bar chart of the labels and every categorical feature



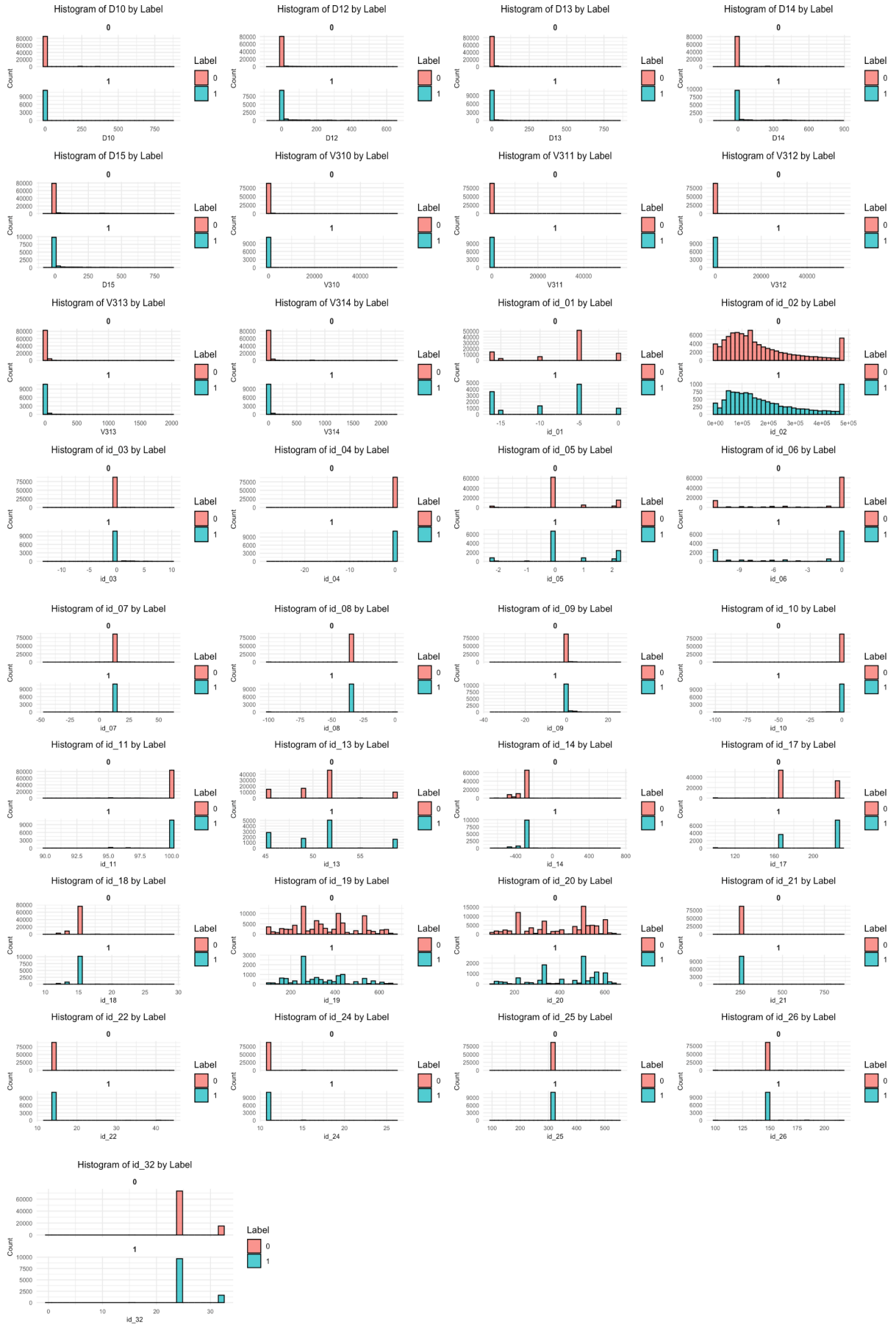
Appendix 7: Table of the OutlierRate for each feature

feature outlierRate			feature outlierRate		
1	TranID	0.00000	34	D12	0.14528
2	TranDT	0.00000	35	D13	0.09195
3	TranDTHour	0.00000	36	D14	0.12210
4	TranAmt	0.06106	37	D15	0.14331
5	addr1	0.31542	38	V310	0.15347
6	addr2	0.02132	39	V311	0.04184
7	dist2	0.26259	40	V312	0.11622
8	card1	0.00000	41	V313	0.11283
9	card2	0.00000	42	V314	0.11774
10	card3	0.00074	43	id_01	0.18235
11	card5	0.00000	44	id_02	0.05990
12	C1	0.11530	45	id_03	0.01808
13	C2	0.17540	46	id_04	0.00449
14	C3	0.01628	47	id_05	0.20951
15	C4	0.26661	48	id_06	0.15243
16	C6	0.23260	49	id_07	0.03432
17	C7	0.06016	50	id_08	0.03400
18	C8	0.11992	51	id_09	0.03317
19	C10	0.11224	52	id_10	0.01426
20	C11	0.24334	53	id_11	0.05411
21	C12	0.08709	54	id_13	0.29443
22	C13	0.12395	55	id_14	0.24385
23	C14	0.31484	56	id_17	0.00000
24	D1	0.20695	57	id_18	0.13611
25	D2	0.21194	58	id_19	0.00000
26	D3	0.19158	59	id_20	0.00000
27	D4	0.16049	60	id_21	0.01823
28	D5	0.10310	61	id_22	0.00314
29	D6	0.18123	62	id_24	0.01329
30	D7	0.11345	63	id_25	0.01834
31	D8	0.26705	64	id_26	0.03563
32	D9	0.48909	65	id_32	0.16769
33	D10	0.03994			

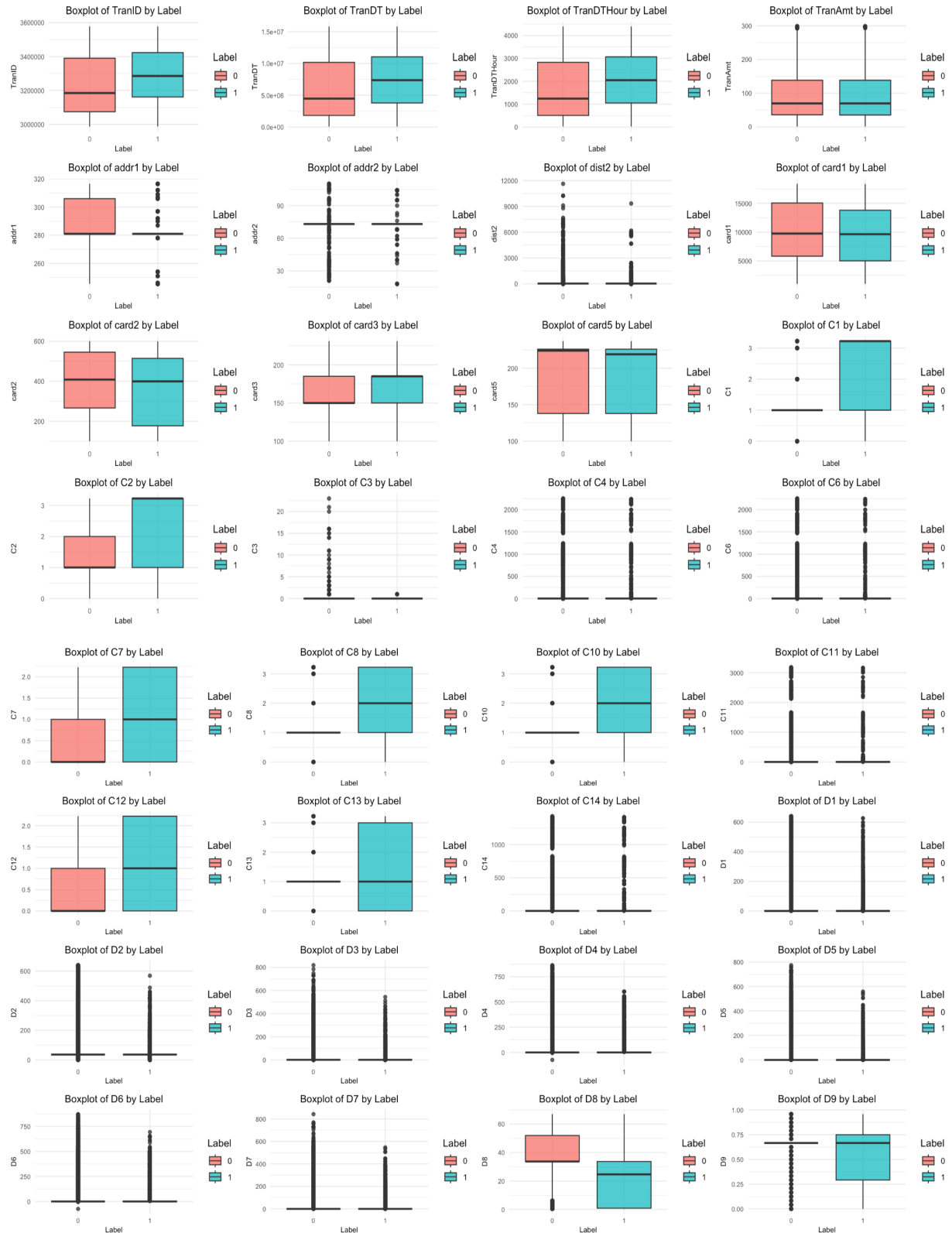
Average outlier rate = 0.1118

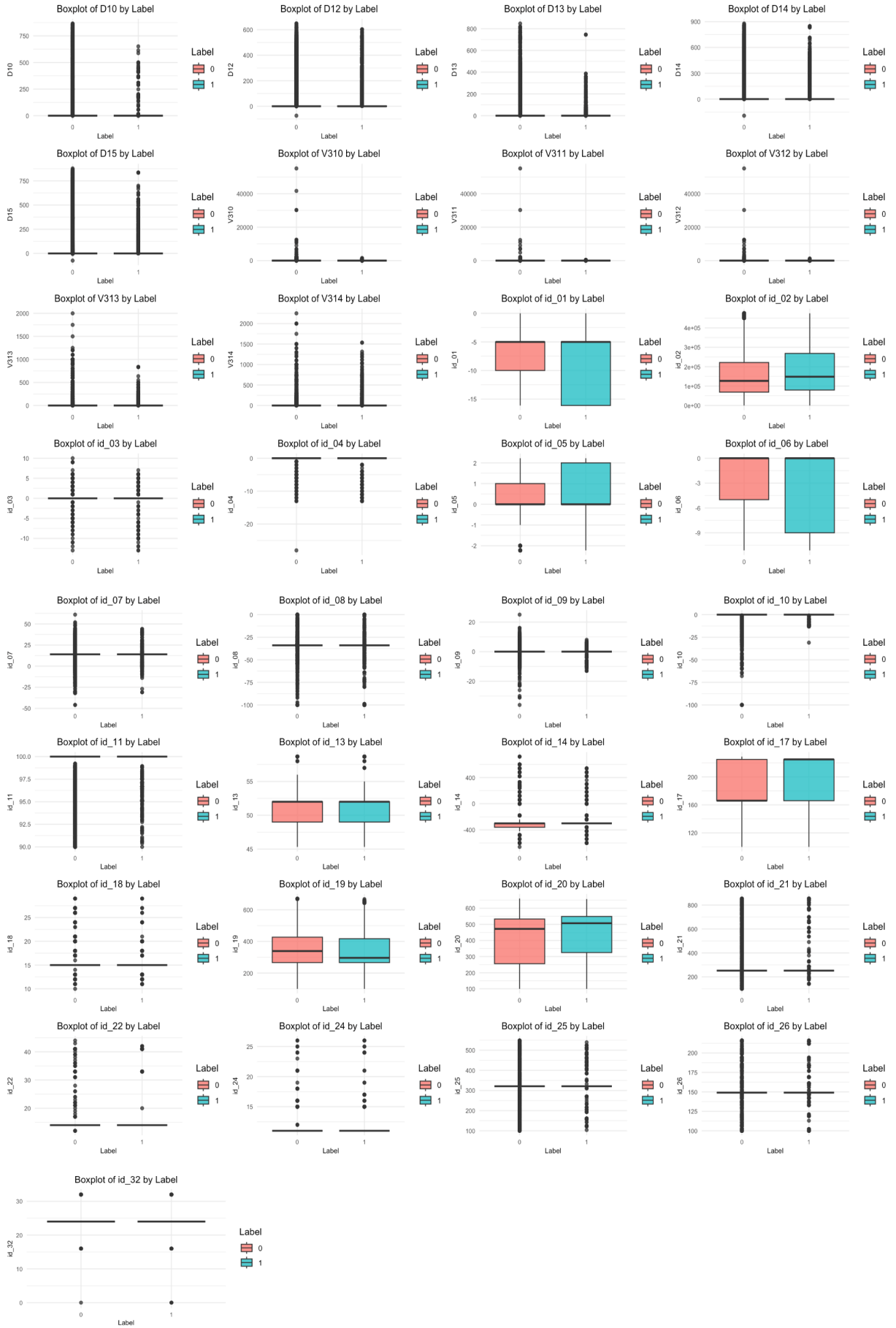
Appendix 8: Bivariate Histogram of Numerical Features by Fraud Label



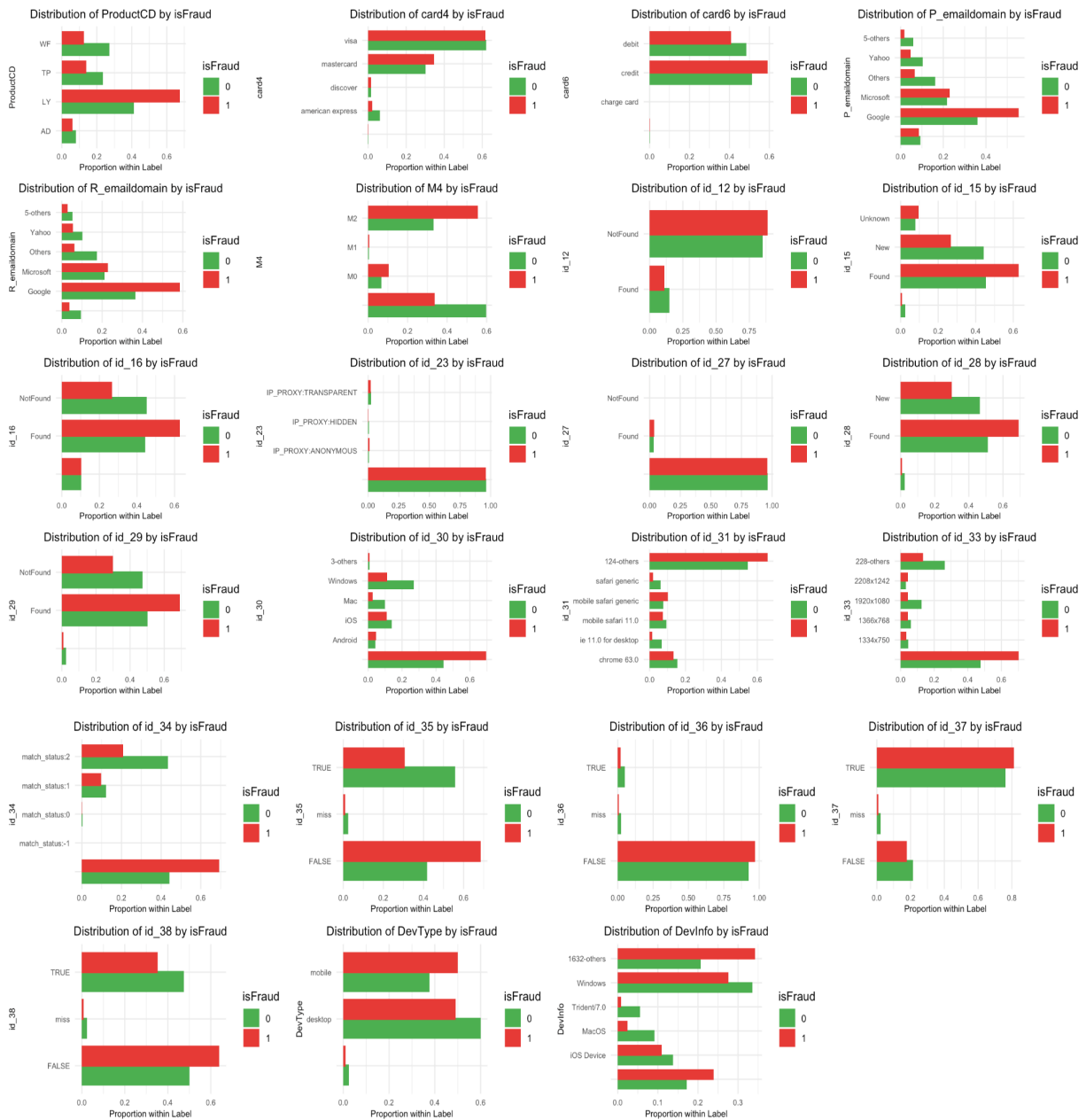


Appendix 9: Bivariate Boxplots of Numerical Features by Fraud Label



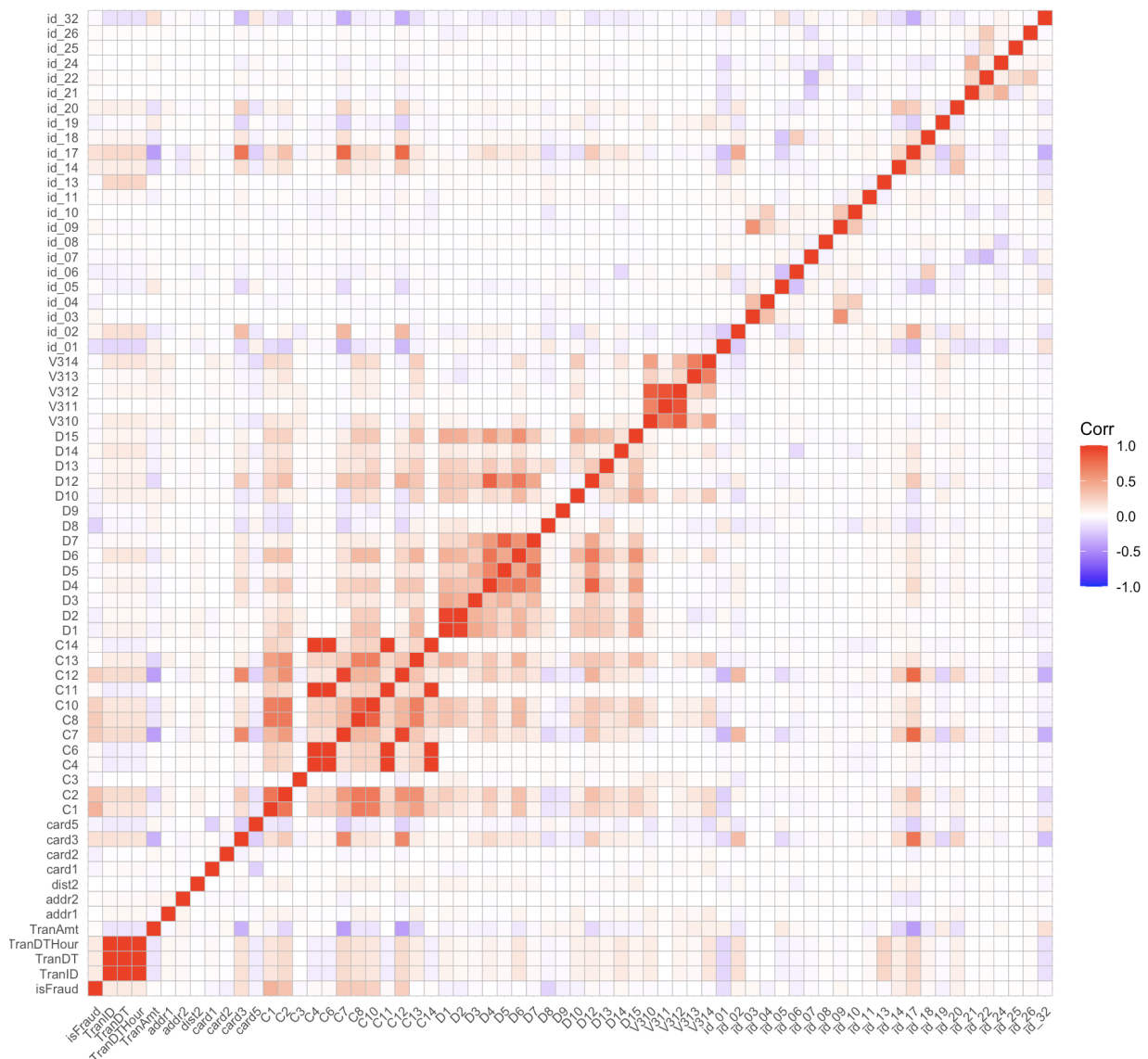


Appendix 10: Bivariate Bar Chart of Categorical Features by Fraud Label



Appendix 11: Correlation Matrix of Numerical Features and Fraud Label

Correlation Matrix of Numerical Features and Fraud Label



Correlation score between isFraud and the numerical features

TranID	TranDT	TranDTHour	TranAmt	addr1	addr2
0.1107075313	0.1065964464	0.1065973526	0.0103690597	-0.0037050543	0.0192088196
dist2	card1	card2	card3	card5	C1
0.0039334887	-0.0403003208	-0.0661493598	0.1735252723	-0.0551078989	0.4223317362
C2	C3	C4	C6	C7	C8
0.3269094342	-0.0294576187	0.0265295837	0.0265036543	0.2771414336	0.2763675048
C10	C11	C12	C13	C14	D1
0.2490529810	0.0337173071	0.2632545859	-0.0096822164	0.0164475768	-0.0514546720
D2	D3	D4	D5	D6	D7
-0.0693500782	-0.0192057718	0.0003184065	-0.0336308916	-0.0104240238	-0.0320201040
D8	D9	D10	D12	D13	D14
-0.1858057542	-0.0672717509	-0.0462324108	0.0254510214	-0.0290141429	0.0357200635
D15	V310	V311	V312	V313	V314
-0.0215326204	-0.0218984120	-0.0054366969	-0.0132936359	-0.0031753094	-0.0158367983
id_01	id_02	id_03	id_04	id_05	id_06
-0.1439816293	0.0623403633	0.0436122411	-0.0574385264	0.0124265097	-0.0703497258
id_07	id_08	id_09	id_10	id_11	id_13
-0.0195994980	-0.0136448489	0.0351384863	0.0019555774	0.0081423423	-0.0230901010
id_14	id_17	id_18	id_19	id_20	id_21
0.0860330744	0.1741218948	0.0278526656	-0.0483364411	0.0698733430	0.0143209344
id_22	id_24	id_25	id_26	id_32	
0.0263045420	0.0025897758	0.0081393400	0.0225034561	-0.0235121795	

Feature with abs(corr) ≥ 0.1 [Selected]

TranID TranDT TranDTHour card3 C1 C2 C7 C8 C10 C12 D8 id_01 id_17

Feature with abs(corr) < 0.1 [Dropped]

TranAmt addr1 addr2 dist2 card1 card2 card5 C3 C4 C6 C11 C13 C14 D1 D2 D3 D4 D5 D6 D7 D9 D10 D12 D13 D14 D15 V310 V311 V312 V313 V314 id_02 id_03 id_04 id_05 id_06 id_07 id_08 id_09 id_10 id_11 id_13 id_14 id_18 id_19 id_20 id_21 id_22 id_24 id_25 id_26 id_32

Appendix 12: Official Dataset Description

- Transaction Table *

-
- TransactionDT: timedelta from a given reference datetime (not an actual timestamp)
 - TransactionAMT: transaction payment amount in USD
 - ProductCD: product code, the product for each transaction
 - card1 - card6: payment card information, such as card type, card category, issue bank, country, etc.
 - addr: address
 - dist: distance
 - P_ and (R_) emaildomain: purchaser and recipient email domain
 - C1-C14: counting, such as how many addresses are found to be associated with the payment card, etc. The actual meaning is masked.
 - D1-D15: timedelta, such as days between previous transaction, etc.
 - M1-M9: match, such as names on card and address, etc.
 - Vxxx: Vesta engineered rich features, including ranking, counting, and other entity relations.

Categorical Features: ProductCD card1 - card6 addr1, addr2 P_emaildomain R_emaildomain M1 - M9

- Identity Table *

Variables in this table are identity information – network connection information (IP, ISP, Proxy, etc) and digital signature (UA/browser/os/version, etc) associated with transactions. They're collected by Vesta's fraud protection system and digital security partners. (The field names are masked and pairwise dictionary will not be provided for privacy protection and contract agreement)

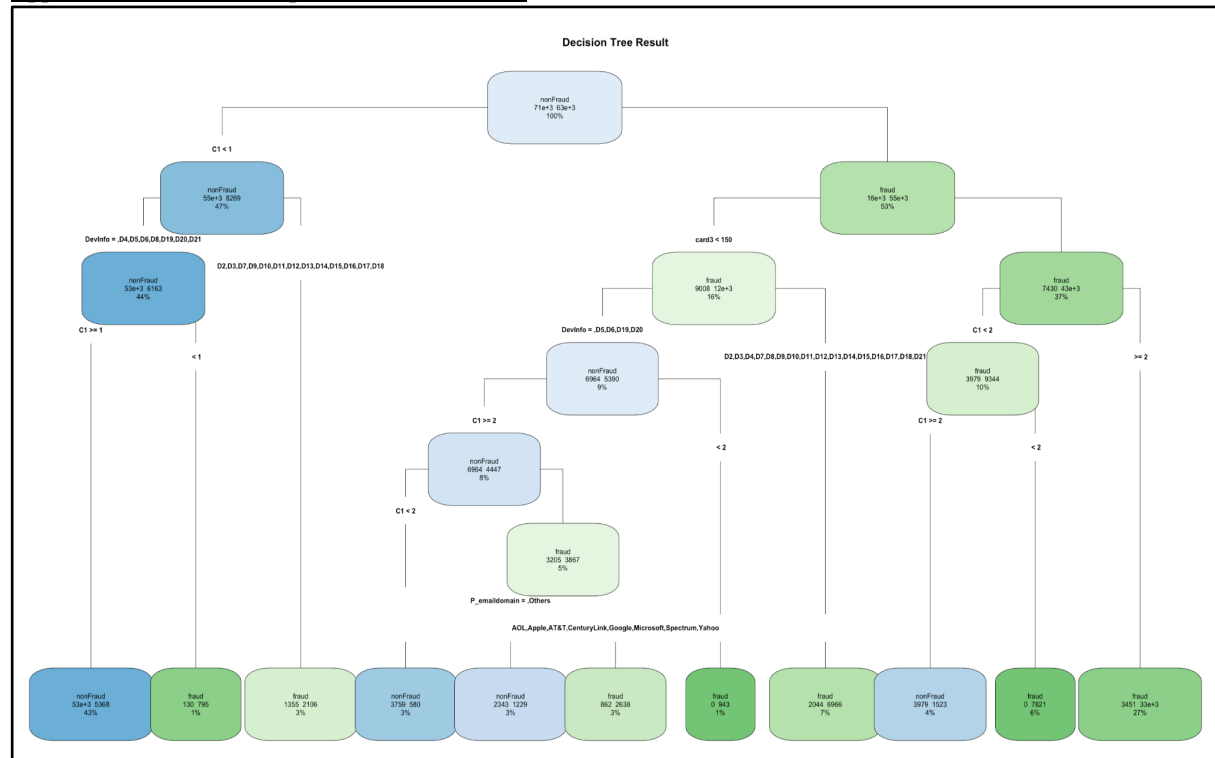
Categorical Features: DeviceType DeviceInfo id_12 - id_38

Source: <https://www.kaggle.com/c/ieee-fraud-detection/discussion/101203>

Identity-related fields [We used]

id_01 id_12 id_15 id_16 id_17 id_23 id_27 id_28 id_29 id_30 id_31 id_33 id_34 id_35 id_36 id_37 id_38

Appendix 13: Decision Graph of the Decision Tree



Note: Mapping Table for Device Information Aliases

[D1] ""	[D2]]"ALE-L23 Build/HuaweiALE-L23"	[D3] "hi6210sf Build/MRA58K"
[D4] "iOS Device"	[D5]]"MacOS"	[D6] "rv:11.0"
[D7] "rv:52.0"	[D8] "rv:57.0"	[D9] "rv:58.0"
[D10] "rv:59.0"	[D11] "SM-A300H Build/LRX22G"	[D12]"SM-G531H Build/LMY48B"
[D13] "SM-G532M Build/MMB29T"	[D14] "SM-G610M Build/MMB29K"	[D15] "SM-G935F Build/NRD90M"
[D16] "SM-G950U Build/NRD90M"	[D17] "SM-G955U Build/NRD90M"	[D18] "SM-J700M Build/MMB29K"
[D19] "Trident/7.0"	[D18] "Windows"	[D21] "1617-others"